# Virtual Web Hosting

Session 16




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# Objectives

- *Define virtual web hosting*
- *Identify the types of virtual hosts*
- *Describe the process of configuring an IP-based virtual host*
- *Describe the process of configuring a name-based virtual host*
- *Describe the process of configuring Apache Web server for mixed virtual hosting*

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# Virtual Web Hosting

- Technique in which Websites with different names run on the same server
- Provides:
  - Domain name registration
  - File storage
  - Directory services for the files from where the Web page is built
- Runs multiple ‘virtual’ Web servers on a single physical host computer
- Allows a single computer to host different independent Websites

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# Virtual Host

- Enables server to respond to different IP address, hostnames, and ports
- Has the capacity to run different servers on the same server
- Configured by using the `httpd.conf` file
- Two types
  - IP-based virtual hosts
    - Uses different IP addresses for each Website
    - Uses multiple IP addresses
  - Name-based virtual hosts
    - Uses names to determine a Website
    - Share a single IP address

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# Directives Used to Create A Virtual Host

- `<VirtualHost>` Directive
- `NameVirtualHost` Directive
- `ServerName` Directive
- `ServerPath` Directive
- `ServerAlias` Directive

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# <VirtualHost> Directive

- Contain instructions that are applicable to specific IP addresses or hostnames
- Apache uses this directive when it receives a request for a document on a particular virtual host set up on the server

**Syntax**

```
<VirtualHost   addr[:port]   [addr[:port]]   ...>   ...
</VirtualHost>
```

**Snippet**

```
<VirtualHost 192.154.127.21:80>
```

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# NameVirtualHost Directive

- Defines the IP address for name-based virtual hosting
- Apache Web server uses this IP address to receive requests for the name-based virtual hosts

**Syntax**

```
NameVirtualHost addr[:port]
```

**Snippet**

```
NameVirtualHost 196.214.370.45:80
```

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# ServerName Directive

- Defines the hostname and port used by the server for identification
- If not specified, then the server determines the hostname from the IP address by performing a reverse lookup

**Syntax**

```
ServerName fully-qualified-domain-name[:port]
```

**Snippet**

```
ServerName www.customer.com
```

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# ServerPath Directive

- Defines the URL pathname for the host to use with a name-based virtual host
- Used only inside a `<VirtualHost>` section in the `httpd.conf` file

**Syntax**

```
ServerPath URL-path
```

**Snippet**

```
<VirtualHost  204.148.170.12>
...
ServerName www.myapacheserver.com
DocumentRoot /www/myapache/htdocs
ServerPath /myapache
...
</VirtualHost>
```

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# ServerAlias Directive

- Defines alternate names for the hosts used in the name-based virtual host
- Must be used along with a `ServerName` directive inside a `<VirtualHost>` section of the `httpd.conf` file

**Syntax**

```
ServerAlias hostname [hostname]...
```

**Snippet**

```
<VirtualHost 198.121.130.12:80>
  ServerName www.customer.com
  ServerAlias customer.com *. customer.com
  ...
</VirtualHost>
```

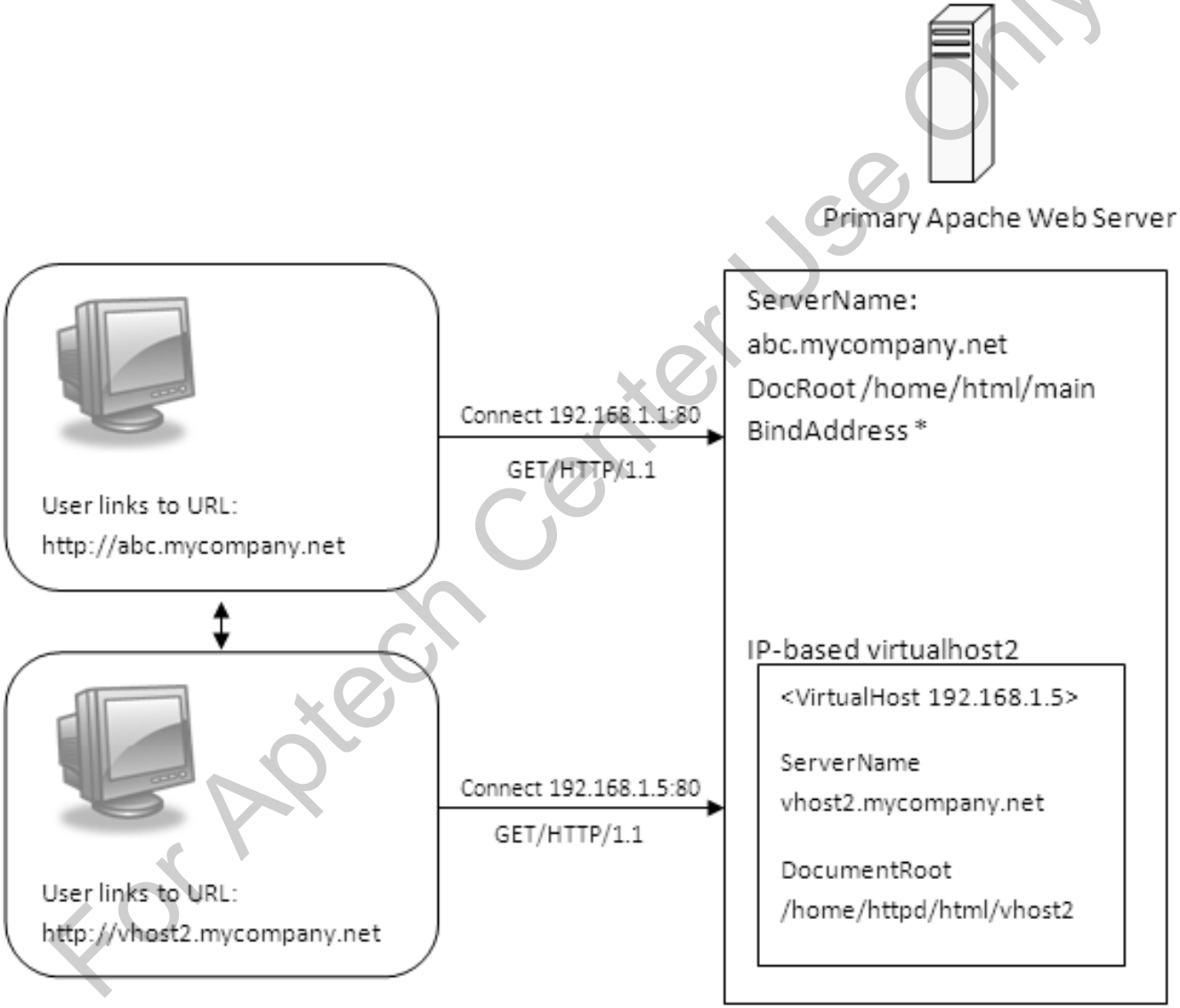
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- Each Website served by Apache server has a unique IP address assigned to it
- Must be assigned multiple IP addresses to the server to support virtual hosting
- Two ways of configuring are as follows:
  - Run a separate `httpd` daemon for each hostname
  - Run a single daemon that supports all the virtual hosts

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# IP-based Virtual Host






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- Steps to set up:
  - Register Domain Name
    - Sets up a virtual host and registers the domain name for the virtual host
  - Request the DNS Addition
    - Registers the domain name and requests the entry of the domain to be added to the domain name records

Snippet

```
<VirtualHost 198.81.129.100:80>
  ServerAdmin webmaster@futurasoftltd.com
  DocumentRoot  /groups/futurasoft/www
  ServerName  www.futurasoftltd.com
  ErrorLog  /groups/futurasoft/logs/error_log
  TransferLog  /groups/futurasoft/logs/access_log
</VirtualHost>
```

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# Name-based Virtual Host

- Use multiple hostnames for the same Web server IP address
- Method of virtual hosting that requires a simple network configuration and no additional hardware or software
- Apache server retrieves the required virtual hostname from the `Host:` header
- The server knows only the IP address of the interface

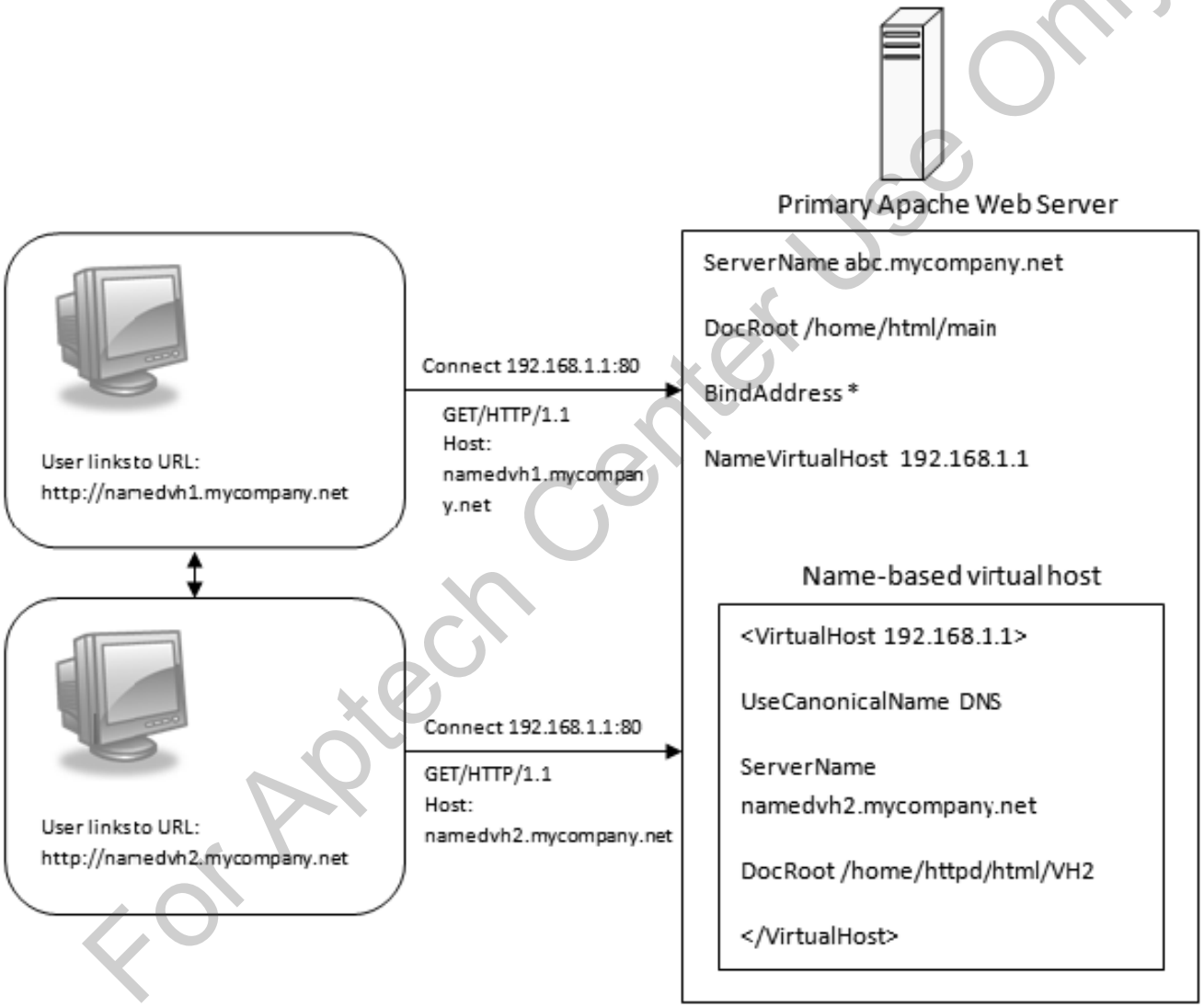
**Snippet**

```
GET / HTTP/1.1 Host: abc.mycompany.net
```

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# Name-based Virtual Host






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- Steps to create are as follows:
  1. Select the IP address and the port
  2. Create a `<VirtualHost>` block in the `httpd.conf` file
  3. Designate the host to be served
  4. Specify the file system in which the contents on the `<VirtualHost>` must be stored

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## Snippet

```
NameVirtualHost 210.145.160.42:80
<VirtualHost 210.145.160.42:80>
  ServerName www.mycompany.com
  DocumentRoot /groups/mycompany/www
  ServerAlias mycompany.com *.mycompany.com
  ServerPath  /mycompany
</VirtualHost>
<VirtualHost 210.145.160.42:443>
  ServerName www.abcompany.com
  DocumentRoot /groups/abcompany/www
  ServerPath /abcompany
</VirtualHost>
```

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# Combining IP-based and Name-based Virtual Hosting

- Name-based and IP-based virtual hosts can be configured in the same server configuration if the name-based and IP-based virtual hosts share a different IP address
- To configure Apache server for name-based and IP-based virtual hosting, enter the code in the `httpd.conf` file

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# Combining IP-based and Name-based Virtual Hosting

```
Listen 80
NameVirtualHost  204.148.170.10
<VirtualHost 204.148.170.10>
  DocumentRoot /www/mycompany1
  ServerName www.mycompany1.com
  ServerAdmin  webmaster@mycompany1.com
</VirtualHost>
<VirtualHost 204.148.170.10>
  DocumentRoot /www/mycompany2
  ServerName www.mycompany2.org
  ServerAdmin webmaster@mycompany2.org
</VirtualHost>
# Creating IP-based virtual hosts
<VirtualHost 204.148.170.15>
  DocumentRoot /www/mycompany5
  ServerName www.mycompany5.com
  ServerAdmin webmaster@mycompany5.com
  ErrorLog /www/logs/mycompany5_error_log
  TransferLog  /www/logs/mycomany5_access_log
</VirtualHost>
<VirtualHost 204.148.170.16>
  DocumentRoot /www/mycompany6
  ServerName www.mycompany6.gov
  ServerAdmin  webmaster@mycompany6.gov
  ErrorLog /www/logs/mycompany6_error_log
  TransferLog  /www/logs/mycompany6_access_log
</VirtualHost>
```

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# Summary

- Virtual Web hosting means running more than one Website on a single machine
- There are two types of virtual hosts that are as follows:
  - IP-based virtual hosts
  - Name-based virtual hosts
- In an IP-based virtual hosting, each Website has its own IP address
- In name-based virtual hosting, the server depends on the client to provide the hostname as a part of the HTTP header
- Apache can be configured to support name-based and IP-based virtual hosting

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